

 INAF ISTITUTO NAZIONALE DI ASTRONOMIA OSSERVATORIO ASTRONOMICO DI ARCETRI	FATE MONTHLY REPORT Doc.No : FATE-001 Version : 1.0 Date : 02-12-2024	 CONSORZIO LaMMA
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FORECAST SYSTEM FOR ATMOSPHERE AND TURBULENCE at the VERY LARGE TELESCOPE

Performance assessment
Annex
October 2024



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1 NIGHT1: Contingency tables

Tables in section 1 are included in the range: 1-22.

wind speed ms^{-1}	Observations		
	ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	2248	905
	4.2 < ws < 7.6	1489	1779
	ws > 7.6	198	688
Number of points 11842 ; PC=65.3%; EBD=2.7%; POD1=57.1%; POD2=52.8%; POD3=81.7%			

Table 1: night1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}	Observations		
	ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	3646	192
	4.2 < ws < 7.6	289	2997
	ws > 7.6	0	183
Number of points 11842 ; PC=92.6%; EBD=0.0%; POD1=92.7%; POD2=88.9%; POD3=95.3%			

Table 2: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

wind speed ms^{-1}	Observations		
	ws < 12.0	12.0 < ws < 18.0	ws > 18.0
Model data	ws < 12.0	10019	126
	12.0 < ws < 18.0	98	1265
	ws > 18.0	0	61
Number of points 11842 ; PC=97.3%; EBD=0.0%; POD1=99.0%; POD2=87.1%; POD3=85.7%			

Table 3: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0)

FATE - Monthly Report

wind direction degree		North	East	South	Observations West
Model data	North	5846	283	0	92
	East	294	522	139	0
	South	34	160	1289	103
	West	267	50	7	103
Number of points 9189 ; PC=84.4%; EBD=0.9%; POD1=90.8%; POD2=51.4%; POD3=89.8%; POD4=34.6%					

Table 4: night1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations West
Model data	North	6353	65	0	21
	East	70	919	38	0
	South	0	31	1390	9
	West	18	0	7	268
Number of points 9189 ; PC=97.2%; EBD=0.0%; POD1=98.6%; POD2=90.5%; POD3=96.9%; POD4=89.9%					

Table 5: night1 - Contingency table for wind direction (degree) processed with AR (1H)

relative humidity percent		Observations		
		rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	3501	1028	9
	7.1 < rh < 15.3	1876	2496	290
	rh > 15.3	185	783	1384
Number of points 11552 ; PC=63.9%; EBD=1.7%; POD1=62.9%; POD2=58.0%; POD3=82.2%				

Table 6: night1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent		Observations		
		rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	5298	226	0
	7.1 < rh < 15.3	264	4016	53
	rh > 15.3	0	65	1630
Number of points 11552 ; PC=94.7%; EBD=0.0%; POD1=95.3%; POD2=93.2%; POD3=96.9%				

Table 7: night1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent		Observations		
		rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	11510	6	0
	50.0 < rh < 70.0	3	33	0
	rh > 70.0	0	0	0
Number of points 11552 ; PC=99.9%; EBD=0.0%; POD1=100.0%; POD2=84.6%; POD3=NaN%				

Table 8: night1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	5647	1012	10
	1.9 < pwv < 3.5	496	2145	377
	pwv > 3.5	21	181	1855
Number of points 11744 ; PC=82.1%; EBD=0.3%; POD1=91.6%; POD2=64.3%; POD3=82.7%				

Table 9: night1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	6069	117	0
	1.9 < pwv < 3.5	95	3188	31
	pwv > 3.5	0	33	2211
Number of points 11744 ; PC=97.6%; EBD=0.0%; POD1=98.5%; POD2=95.5%; POD3=98.6%				

Table 10: night1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

FATE - Monthly Report

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	1033	1020	289
	0.6 < see < 0.8	754	983	886
	see > 0.8	82	318	810
Number of points 6175 ; PC=45.8%; EBD=6.0%; POD1=55.3%; POD2=42.4%; POD3=40.8%				

Table 11: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	1843	382	5
	0.6 < see < 0.8	268	1742	171
	see > 0.8	4	245	1897
Number of points 6175 ; PC=83.6%; EBD=0.1%; POD1=87.1%; POD2=73.5%; POD3=91.5%				

Table 12: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	1801	0	191
	0.6 < see < 0.8	0	2259	191
	see > 0.8	68	62	1794
Number of points 6175 ; PC=94.8%; EBD=4.2%; POD1=96.4%; POD2=97.3%; POD3=82.4%				

Table 13: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	2111	4	5
	0.6 < see < 0.8	0	2358	5
	see > 0.8	4	7	2068
Number of points 6175 ; PC=99.7%; EBD=0.1%; POD1=99.8%; POD2=99.5%; POD3=99.5%				

Table 14: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.24

FATE - Monthly Report

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1064	644	45
	$3.4 < \tau < 5.8$	363	1285	515
	$\tau > 5.8$	27	441	1066
Number of points 5450 ; PC=62.7%; EBD=1.3%; POD1=73.2%; POD2=54.2%; POD3=65.6%				

Table 15: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1357	149	1
	$3.4 < \tau < 5.8$	97	2080	162
	$\tau > 5.8$	0	141	1463
Number of points 5450 ; PC=89.9%; EBD=0.0%; POD1=93.3%; POD2=87.8%; POD3=90.0%				

Table 16: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1345	135	205
	$3.4 < \tau < 5.8$	82	2063	205
	$\tau > 5.8$	27	172	1421
Number of points 5450 ; PC=88.6%; EBD=4.3%; POD1=92.5%; POD2=87.0%; POD3=77.6%				

Table 17: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1454	0	12
	$3.4 < \tau < 5.8$	0	2359	12
	$\tau > 5.8$	0	11	1614
Number of points 5450 ; PC=99.6%; EBD=0.2%; POD1=100.0%; POD2=99.5%; POD3=98.5%				

Table 18: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 1.22

FATE - Monthly Report

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	967	1041
	0.6 < glf < 0.7	528	746
	glf > 0.7	77	331
Number of points 5573 ; PC=38.7%; EBD=10.6%; POD1=61.5%; POD2=35.2%; POD3=23.4%			

Table 19: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1323	292
	0.6 < glf < 0.7	236	1556
	glf > 0.7	13	270
Number of points 5573 ; PC=79.5%; EBD=0.4%; POD1=84.2%; POD2=73.5%; POD3=82.3%			

Table 20: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1483	242
	0.6 < glf < 0.7	12	1858
	glf > 0.7	77	18
Number of points 5573 ; PC=83.5%; EBD=11.6%; POD1=94.3%; POD2=87.7%; POD3=55.1%			

Table 21: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1552	18
	0.6 < glf < 0.7	7	2099
	glf > 0.7	13	1
Number of points 5573 ; PC=99.2%; EBD=0.4%; POD1=98.7%; POD2=99.1%; POD3=99.2%			

Table 22: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

2 DAY1: Contingency tables

Tables in section 2 are included in the range: 23-32.

wind speed ms^{-1}		Observations		
		ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	2211	611	113
	4.6 < ws < 7.4	1777	2250	689
	ws > 7.4	158	803	3633
Number of points 12245 ; PC=66.1%; EBD=2.2%; POD1=53.3%; POD2=61.4%; POD3=81.9%				

Table 23: day1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed		Observations		
ms^{-1}		ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	3686	265	4
	4.6 < ws < 7.4	458	3191	177
	ws > 7.4	2	208	4254
Number of points 12245 ; PC=90.9%; EBD=0.0%; POD1=88.9%; POD2=87.1%; POD3=95.9%				

Table 24: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

wind speed		Observations		
ms^{-1}		ws < 12.0	12.0 < ws < 18.0	ws > 18.0
Model data	ws < 12.0	9912	151	0
	12.0 < ws < 18.0	109	1519	67
	ws > 18.0	0	64	423
Number of points 12245 ; PC=96.8%; EBD=0.0%; POD1=98.9%; POD2=87.6%; POD3=86.3%				

Table 25: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0)

FATE - Monthly Report

wind direction degree		North	East	South	Observations	West
Model data	North	4932	367	0		118
	East	74	337	46		0
	South	25	17	812		303
	West	1290	24	108		2158
Number of points 10611 ; PC=77.6%; EBD=0.5%; POD1=78.0%; POD2=45.2%; POD3=84.1%; POD4=83.7%						

Table 26: day1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations	West
Model data	North	6137	50	2		114
	East	49	688	8		0
	South	0	5	927		26
	West	135	2	29		2439
Number of points 10611 ; PC=96.0%; EBD=0.0%; POD1=97.1%; POD2=92.3%; POD3=96.0%; POD4=94.6%						

Table 27: day1 - Contingency table for wind direction (degree) processed with AR (1H)

relative humidity percent		Observations		
		rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	4367	1036	8
	8.9 < rh < 17.9	1740	2853	448
	rh > 17.9	71	482	1184
Number of points 12189 ; PC=68.9%; EBD=0.6%; POD1=70.7%; POD2=65.3%; POD3=72.2%				

Table 28: day1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent		Observations		
		rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	5931	235	0
	8.9 < rh < 17.9	247	4061	68
	rh > 17.9	0	75	1572
Number of points 12189 ; PC=94.9%; EBD=0.0%; POD1=96.0%; POD2=92.9%; POD3=95.9%				

Table 29: day1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent		Observations		
		rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	12109	10	0
	50.0 < rh < 70.0	7	59	4
	rh > 70.0	0	0	0
Number of points 12189 ; PC=99.8%; EBD=0.0%; POD1=99.9%; POD2=85.5%; POD3=0.0%				

Table 30: day1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

precipitable water vapor mm		Observations		
		pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	6070	669	0
	2.1 < pwv < 3.7	690	2145	474
	pwv > 3.7	40	254	1821
Number of points 12163 ; PC=82.5%; EBD=0.3%; POD1=89.3%; POD2=69.9%; POD3=79.3%				

Table 31: day1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	6686	55	0
	2.1 < pwv < 3.7	114	2971	40
	pwv > 3.7	0	42	2255
Number of points 12163 ; PC=97.9%; EBD=0.0%; POD1=98.3%; POD2=96.8%; POD3=98.3%				

Table 32: day1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

3 NIGHT2: Contingency tables

Tables in section 3 are included in the range: 33-42.

wind speed ms^{-1}	Observations		
	ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	2453	994
	4.2 < ws < 7.6	1430	1688
	ws > 7.6	265	745
Number of points 12127 ; PC=64.5%; EBD=3.2%; POD1=59.1%; POD2=49.3%; POD3=81.0%			

Table 33: night2 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	5930	293	16
	East	289	544	157
	South	57	148	1230
	West	253	59	13

Number of points 9322 ; PC=84.3%; EBD=1.4%; POD1=90.8%; POD2=52.1%; POD3=86.9%; POD4=46.5%

Table 34: night2 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	3463	994
	7.1 < rh < 15.3	2056	2568
	rh > 15.3	210	845

Number of points 11823 ; PC=63.1%; EBD=1.8%; POD1=60.4%; POD2=58.3%; POD3=84.6%

Table 35: night2 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	5695	1080
	1.9 < pwv < 3.5	605	2201
	pwv > 3.5	25	234

Number of points 12017 ; PC=79.1%; EBD=0.5%; POD1=90.0%; POD2=62.6%; POD3=74.1%

Table 36: night2 - Contingency table for precipitable water vapor (mm) in standard configuration

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	1268	1231	411
	0.6 < see < 0.8	942	1285	1076
	see > 0.8	142	466	1518
Number of points 8339 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 37: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	2244	0	276
	0.6 < see < 0.8	0	2860	276
	see > 0.8	108	122	2729
Number of points 8339 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 38: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1944	947	91
	$3.4 < \tau < 5.8$	446	1525	647
	$\tau > 5.8$	42	559	1254
Number of points 7455 ; PC=63.4%; EBD=1.8%; POD1=79.9%; POD2=50.3%; POD3=63.0%				

Table 39: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	2280	200	321
	$3.4 < \tau < 5.8$	110	2603	321
	$\tau > 5.8$	42	228	1671
Number of points 7455 ; PC=87.9%; EBD=4.9%; POD1=93.8%; POD2=85.9%; POD3=72.2%				

Table 40: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1156	1305
	0.6 < glf < 0.7	942	1300
	glf > 0.7	141	352
Number of points 7644 ; PC=40.5%; EBD=10.9%; POD1=51.6%; POD2=44.0%; POD3=26.0%			

Table 41: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	2066	373
	0.6 < glf < 0.7	32	2548
	glf > 0.7	141	36
Number of points 7644 ; PC=82.2%; EBD=12.0%; POD1=92.3%; POD2=86.2%; POD3=54.1%			

Table 42: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

4 DAY2: Contingency tables

Tables in section 4 are included in the range: 43-46.

wind speed ms^{-1}		Observations		
		ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	2217	603	155
	4.6 < ws < 7.4	1783	2271	651
	ws > 7.4	162	790	3629
Number of points 12261 ; PC=66.2%; EBD=2.6%; POD1=53.3%; POD2=62.0%; POD3=81.8%				

Table 43: day2 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	4953	433	13
	East	85	254	45
	South	36	41	745
	West	1247	17	162

Number of points 10611 ; PC=76.5%; EBD=0.6%; POD1=78.4%; POD2=34.1%; POD3=77.2%; POD4=84.1%

Table 44: day2 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	4066	887
	8.9 < rh < 17.9	2008	2882
	rh > 17.9	104	602

Number of points 12189 ; PC=66.5%; EBD=1.0%; POD1=65.8%; POD2=65.9%; POD3=70.4%

Table 45: day2 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	5852	693
	2.1 < pwv < 3.7	905	2124
	pwv > 3.7	43	251

Number of points 12163 ; PC=79.3%; EBD=0.4%; POD1=86.1%; POD2=69.2%; POD3=72.5%

Table 46: day2 - Contingency table for precipitable water vapor (mm) in standard configuration

5 NIGHT3: Contingency tables

Tables in section 5 are included in the range: 47-56.

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	2018	1044	153
	4.2 < ws < 7.6	1699	1470	730
	ws > 7.6	320	899	3660
Number of points 11993 ; PC=59.6%; EBD=3.9%; POD1=50.0%; POD2=43.1%; POD3=80.6%				

Table 47: night3 - Contingency table for wind speed (ms^{-1}) in standard configuration

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	5807	293	16
	East	410	502	204
	South	156	217	1093
	West	197	39	13

Number of points 9260 ; PC=80.9%; EBD=2.3%; POD1=88.4%; POD2=47.8%; POD3=82.4%; POD4=28.1%

Table 48: night3 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	3247	968
	7.1 < rh < 15.3	2136	2556
	rh > 15.3	268	854

Number of points 11716 ; PC=61.0%; EBD=2.5%; POD1=57.5%; POD2=58.4%; POD3=79.7%

Table 49: night3 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	5648	1173
	1.9 < pwv < 3.5	521	1910
	pwv > 3.5	56	377

Number of points 11832 ; PC=75.1%; EBD=0.8%; POD1=90.7%; POD2=55.2%; POD3=61.6%

Table 50: night3 - Contingency table for precipitable water vapor (mm) in standard configuration

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	1254	1063	391
	0.6 < see < 0.8	871	1266	990
	see > 0.8	222	617	1562
Number of points 8236 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 51: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	2184	0	229
	0.6 < see < 0.8	0	2826	229
	see > 0.8	163	120	2714
Number of points 8236 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%				

Table 52: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1871	923	42
	$3.4 < \tau < 5.8$	524	1567	715
	$\tau > 5.8$	34	497	1191
Number of points 7364 ; PC=62.9%; EBD=1.0%; POD1=77.0%; POD2=52.5%; POD3=61.1%				

Table 53: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	2213	168	331
	$3.4 < \tau < 5.8$	182	2567	331
	$\tau > 5.8$	34	252	1617
Number of points 7364 ; PC=86.9%; EBD=5.0%; POD1=91.1%; POD2=85.9%; POD3=71.0%				

Table 54: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1106	1163
	0.6 < glf < 0.7	947	1372
	glf > 0.7	154	385
Number of points 7553 ; PC=42.3%; EBD=10.3%; POD1=50.1%; POD2=47.0%; POD3=29.7%			

Table 55: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	2013	296
	0.6 < glf < 0.7	40	2594
	glf > 0.7	154	30
Number of points 7553 ; PC=84.1%; EBD=11.0%; POD1=91.2%; POD2=88.8%; POD3=58.6%			

Table 56: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

6 DAY3: Contingency tables

Tables in section 6 are included in the range: 57-60.

wind speed ms^{-1}	Observations		
	ws < 4.6	4.6 < ws < 7.4	ws > 7.4
Model data	ws < 4.6	2060	589
	4.6 < ws < 7.4	1890	2164
	ws > 7.4	212	911
Number of points 12261 ; PC=64.1%; EBD=3.0%; POD1=49.5%; POD2=59.1%; POD3=82.0%			

Table 57: day3 - Contingency table for wind speed (ms^{-1}) in standard configuration

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wind direction degree	Observations			
	North	East	South	West
Model data	North	4827	436	20
	East	82	248	38
	South	87	34	730
	West	1325	27	177

Number of points 10611 ; PC=74.3%; EBD=1.3%; POD1=76.4%; POD2=33.3%; POD3=75.6%; POD4=80.8%

Table 58: day3 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 8.9	8.9 < rh < 17.9	rh > 17.9
Model data	rh < 8.9	3797	762
	8.9 < rh < 17.9	2185	2893
	rh > 17.9	196	716

Number of points 12189 ; PC=64.4%; EBD=1.6%; POD1=61.5%; POD2=66.2%; POD3=70.8%

Table 59: day3 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 2.1	2.1 < pwv < 3.7	pwv > 3.7
Model data	pwv < 2.1	5718	815
	2.1 < pwv < 3.7	1017	1940
	pwv > 3.7	65	313

Number of points 12163 ; PC=76.7%; EBD=0.6%; POD1=84.1%; POD2=63.2%; POD3=72.7%

Table 60: day3 - Contingency table for precipitable water vapor (mm) in standard configuration